



M-Tech Production & Marketing

Technology | Training | Marketing | Projects

Week 2 - Day 3 Report

Internship Program — Batch 2026

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1. Introduction

This report summarizes the activities completed during the M-Tech Production & Marketing AI/ML Internship Program on 8th July 2026. Today I attended an online session and studied Experiment 05 from the Artificial Intelligence & Machine Learning Lab Manual. This experiment focused on Supervised Learning, covering both Regression and Classification algorithms. I learned how to build models that predict continuous values using Regression and models that predict categories using Classification. I also studied core algorithms like Linear Regression, Logistic Regression, Decision Trees, K-Nearest Neighbors, and Support Vector Machines, along with evaluation metrics such as R^2 , RMSE, and Accuracy.

2. Objectives

- Understand the difference between Regression and Classification in Supervised Learning
 - Build Regression models to predict continuous values like prices and scores
 - Build Classification models to predict categories like spam or not spam
 - Learn core supervised algorithms: Linear Regression, Logistic Regression, Decision Tree, KNN, and SVM
 - Split data into training and test sets for proper evaluation
 - Interpret model metrics such as R^2 , RMSE, and Accuracy
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3. Task Description

Today's work involved studying Experiment 05 from the AI/ML Lab Manual. The experiment introduced Supervised Learning in detail, focusing on two main types: Regression and Classification. I learned how Regression is used to predict continuous numerical values such as house prices, temperatures, and exam scores, while Classification is used to predict discrete categories such as spam detection, disease diagnosis, and image recognition. I also studied five important supervised algorithms: Linear Regression, Logistic Regression, Decision Trees, K-Nearest Neighbors (KNN), and Support Vector Machines (SVM). Additionally, I learned about evaluation metrics including R^2 score, RMSE for regression, and Accuracy for classification. I also understood the importance of splitting data into training and test sets to evaluate model performance on unseen data.

4. Work Done

During today's online session and study, I carefully completed Experiment 05 from the lab manual. I started by understanding the difference between Regression and Classification. I learned that Regression predicts continuous numbers like price or temperature, while Classification predicts categories like spam or not spam, or disease vs healthy.

I then studied Linear Regression in detail. I understood that Linear Regression fits a straight line through the data to predict continuous values. I learned how to use the scikit-learn library to build a Linear Regression model, split data into training and test sets using `train_test_split`, and

evaluate the model using R^2 score and RMSE. I learned that R^2 close to 1 means the model explains the data well, and lower RMSE means smaller prediction errors.

Next, I studied Classification and learned about four important algorithms. I understood that Logistic Regression predicts class probabilities using an S-shaped curve, Decision Trees ask a series of yes/no questions to make predictions, K-Nearest Neighbors predicts based on the vote of the k closest data points, and Support Vector Machines find the widest gap between classes to separate them.

I then practiced building a KNN classifier on the Iris dataset and achieved perfect accuracy. I also trained a Logistic Regression and a Decision Tree classifier and compared their accuracies. I learned how to import datasets from scikit-learn, split them, train models, and evaluate them using `accuracy_score` and the `score` method.

Finally, I understood the importance of testing models on data they have not seen during training. This helps us measure real-world performance and avoid overconfidence in our models. I also learned about feature importance in Decision Trees and how it helps us understand which features matter most for predictions.

5. Challenges Faced

The concepts covered in Experiment 05 were well structured and easy to follow. However, understanding the mathematical intuition behind Logistic Regression and the difference between R^2 and RMSE required some extra attention. I also had to spend time understanding how KNN makes predictions using distance calculations and why choosing the right value of k is important. By reviewing the examples and practicing with different datasets, I gained a much clearer understanding of these concepts. The hands-on practice of building multiple models and comparing their performance was very helpful.

6. Key Learnings

Today's work significantly strengthened my understanding of Supervised Learning. I learned that Regression predicts continuous values while Classification predicts categories. I understood the five core supervised algorithms: Linear Regression, Logistic Regression, Decision Tree, KNN, and SVM, and when to use each one. I learned how to evaluate regression models using R^2 and RMSE, and classification models using Accuracy. I also understood the importance of train-test split for proper model evaluation. These concepts provide the essential building blocks for all future machine learning experiments and real-world applications.

7. Conclusion

Studying Experiment 05 provided a solid theoretical and practical foundation in Supervised Learning. I now understand the difference between Regression and Classification, and I can build

and evaluate models for both types of problems. I have hands-on experience with five important supervised algorithms and understand how to choose the right algorithm for a given problem. The knowledge gained from this experiment has improved my confidence in applying supervised learning to real-world problems. This learning will serve as a strong base for upcoming experiments on Unsupervised Learning, Ensemble Methods, and more advanced AI applications throughout the internship.

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Name:**Date:**

08 July 2026

Designation: